Assistive Handwriting Haptic Mechanism Using Deep Learning Speech Recognition

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Abstract. Haptic devices become widespread in education and training. Learning to handwrite for visually impaired people is a difficult task. Using assistive haptic device during handwriting learning for people having a visual impairment is the objective of this work. They don't need anyone to help them in this process. By giving speech command, they can make the mechanism draw the number between any 0-9. Convolutional Neural Network(CNN) is used for speech recognition. Speech recognition for this study is limited only in numbers(0-9)and "unknown" which may occur due to network's inability or given speech command except for a trained number . This paper focuses on dynamic analysis and control of 5R mechanism by using deep learning for speech recognition.

Keywords: Planer 5R Mechanism, Convolutional Neural Network, Haptic Assitive Device, Speech Recognition, Learning Handwriting

1 Introduction

The word haptic stems from Ancient Greek (haptikós, "able to come in contact with") and (háptō, "to touch").In other words, haptics is anything involving touch[1]. Haptics technology emerges as different applications such as alleviation problem for blind people [9], education [6], training [1], medical field [4], computer-aided design [8].This study deals with the application of assistive handwriting for visually impaired people. Since many years, visually impaired people cannot reach educational opportunities as much as normal people. This study enables them to learn how to write on their own by using a speech recognition command. An example prototype of handwriting assistive haptic 5R device is taken from the work(Dede & Kiper) and illustrated in Fig1.In dynamic analysis and control part for the sake of simplicity, the gripper isn't taken into consideration in the calculations. Calculations are based on the model shown in Fig.2.
For the speech recognition part, Convolutional Neural Network (CNN) is used and trained in Matlab and connected to Simulink to provide 5R with the desired trajectory based on speech command.

2 Kinematic Analysis

Each link of 5R can be defined by a fixed vector as shown in Fig.2.Let us define vector $\overrightarrow{O_A A}$ for link2, $\overrightarrow{AP}$ for link3, $\overrightarrow{EP}$ for link4, $\overrightarrow{O_E E}$ for

[1] https://en.wiktionary.org/wiki/haptic

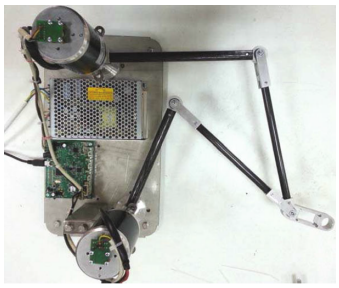

Fig. 1. Handwriting assistive haptic device

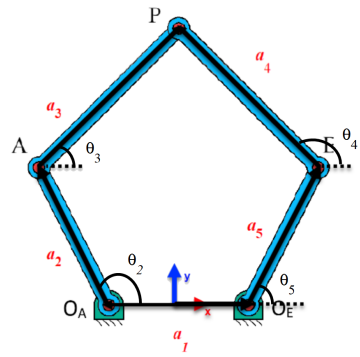

Fig. 2. 5R Model

link5 and $\overrightarrow{O_A O_E}$ for link1.Except $\overrightarrow{O_A O_E}$, the other vectors will be the function of time since the mechanism contains only revolute joints.

2.1 Position Analysis

For position analysis, loop closure equations of vectors on Fig.2 can be written by using Euler equations as below.

$$\frac{-a_1}{2} + a_2 \cdot e^{i\theta_2} + a_3 \cdot e^{i\theta_3} = \frac{a_1}{2} + a_5 \cdot e^{i\theta_{15}} + a_4 \cdot e^{i\theta_{14}} \tag{1}$$

Reel part Re$\{Eq.(1)\}$ and imaginary part Im$\{Eq.(1)\}$

$$\begin{aligned} &\frac{-a_1}{2} + a_2 \cdot \cos(\theta_2) + a_3 \cdot \cos(\theta_3) = \frac{a_1}{2} + a_5 \cdot \cos(\theta_5) + a_4 \cdot \cos(\theta_4) \\ &a_2 \cdot \sin(\theta_2) + a_3 \cdot \sin(\theta_3) = a_5 \cdot \sin(\theta_5) + a_4 \cdot \sin(\theta_4) \end{aligned} \tag{2}$$

Eq.2 is arranged as below and taking square of both side gives us the form shown in Eq.3.
$(a_4 \cdot \cos(\theta_4))^2 = (a_2 \cdot \cos(\theta_2) + a_3 \cdot \cos(\theta_3) - a_5 \cdot \cos(\theta_5) - a_1)^2$

$(a_4 \cdot \sin(\theta_4))^2 = (a_2 \cdot \sin(\theta_2) + a_3 \cdot \sin(\theta_3) - a_5 \cdot \sin(\theta_5))^2$

$$A \cdot \cos(\theta) + B \cdot \sin(\theta) = C \tag{3}$$

$A = 2a_2a_3\cos(\theta_2) - 2a_3a_5\cos(\theta_5) - 2a_3a_1$

$B = 2a_2a_3\sin(\theta_2) - 2a_3a_5\sin(\theta_5)$

$C = -a_1^2 - a_2^2 - a_3^2 - a_5^2 + a_4^2 + 2a_2a_5\cos(\theta_5)\cos(\theta_2) + 2a_1a_2\cos(\theta_2) + 2a_2a_5\sin(\theta_2)\sin(\theta_5) - 2a_1a_5\cos(\theta_5)$

Solution of Eq.(3) is given in Eq.(4).

$$
\begin{aligned}
&t_{1,2} = \frac{B \pm \sqrt{A^2 + B^2 - C^2}}{A + C} \\
&\theta_3 = 2atan(t_1) \qquad or \qquad \theta_3 = 2atan(t_2) \\
&\theta_4 = atan2(a_2 \sin(\theta_2 + a_2 \sin(\theta_3) - a_5 \sin(\theta_5), -a_1 + a_2 \cos(\theta_2) + a_3 \cos(\theta_3) - a_5 \cos(\theta_5))
\end{aligned} \tag{4}
$$

End Effector position can be expressed 2 different ways shown in Eq.(5).

$$
\begin{aligned}
&x = \frac{-a_1}{2} + a_2 \cos(\theta_2) + a_3 \cos(\theta_3) \qquad or \qquad x = \frac{a_1}{2} + a_5 \cos(\theta_5) + a_4 \cos(\theta_4) \\
&y = a_2 sin(\theta_2) + a_3 \cos(\theta_3) \qquad or \qquad y = a_5 \sin(\theta_5) + a_4 \sin(\theta_4)
\end{aligned} \tag{5}
$$

2.2 Velocity Analysis

As an initial value, $\dot{\theta_2}$ and $\dot{\theta_5}$ are given.
Taking derivative both Eq.(2) with respect to time, we can obtain Eq.(6).

$$
\begin{aligned}
&-a_2 \sin(\theta_2)\dot{\theta_2} + a_5 \sin(\theta_5)\dot{\theta_5} - a_3 \sin(\theta_3)\dot{\theta_3} + a_4 \sin(\theta_4)\dot{\theta_4} = 0 \\
&a_2 \cos(\theta_2)\dot{\theta_2} - a_5 \cos(\theta_5)\dot{\theta_5} + a_3 \cos(\theta_3)\dot{\theta_3} - a_4 cos(\theta_4)\dot{\theta_4} = 0
\end{aligned} \tag{6}
$$

The matrix form is given in Eq.7.

$$
\begin{bmatrix} -a_3 \sin(\theta_3) & a_4 \sin(\theta_4) \\ a_3 \cos(\theta_3) & -a_4 \cos(\theta_4) \end{bmatrix} \cdot \begin{bmatrix} \dot{\theta_3} \\ \dot{\theta_4} \end{bmatrix} = \begin{bmatrix} a_2 \sin(\theta_2)\dot{\theta_2} - a_5 \sin(\theta_5)\dot{\theta_5} \\ -a_2 \cos(\theta_2)\dot{\theta_2} + a_5 \cos(\theta_5)\dot{\theta_5} \end{bmatrix} \tag{7}
$$

2.3 Acceleration Analysis

As an initial value, $\ddot{\theta_2}$ and $\ddot{\theta_5}$ are given.
Taking derivative both Eq. (6) with respect to time, we can obtain Eq.(8).

$$
\begin{aligned}
&-a_2 \sin(\theta_2)\ddot{\theta_2} + a_2 \cos(\theta_2)\dot{\theta_2}^2 + a_5 \sin(\theta_5)\ddot{\theta_5} + a_5 \cos(\theta_5)\ddot{\theta_5} - a_3 \sin(\theta_3)\ddot{\theta_3} \\
&-a_3 \cos(\theta_3)\dot{\theta_3}^2 + a_4 \sin(\theta_4)\ddot{\theta_4} + a_4 \cos(\theta_4)\dot{\theta_4}^2 = 0 \\
&a_2 \cos(\theta_2)\ddot{\theta_2} - a_2 \sin(\theta_2)\dot{\theta_2}^2 - a_5 \cos(\theta_5)\ddot{\theta_5} + a_5 \sin(\theta_5)\ddot{\theta_5} + a_3 \cos(\theta_3)\ddot{\theta_3} \\
&-a_3 \sin(\theta_3)\dot{\theta_3}^2 - a_4 \cos(\theta_4)\ddot{\theta_4} + a_4 \sin(\theta_4)\dot{\theta_4}^2 = 0
\end{aligned} \tag{8}
$$

The matrix form is given in Eq.9.

$$
\begin{bmatrix} -a_3 \sin(\theta_3) & a_4 \sin(\theta_4) \\ a_3 \cos(\theta_3) & -a_4 \cos(\theta_4) \end{bmatrix} \cdot \begin{bmatrix} \ddot{\theta_3} \\ \ddot{\theta_4} \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} \tag{9}
$$

where
$A_1 = a_2 \sin(\theta_2)\ddot{\theta_2} - a_2 \cos(\theta_2)\dot{\theta_2}^2 - a_5 \sin(\theta_5)\ddot{\theta_5} - a_5 \cos(\theta_5)\ddot{\theta_5} + a_3 \cos(\theta_3)\dot{\theta_3}^2 - a_4 \cos(\theta_4)\dot{\theta_4}^2$
$A_2 = -a_2 \cos(\theta_2)\ddot{\theta_2} + a_2 \sin(\theta_2)\dot{\theta_2}^2 + a_5 \cos(\theta_5)\ddot{\theta_5} - a_5 \sin(\theta_5)\ddot{\theta_5} + a_3 \sin(\theta_3)\dot{\theta_3}^2 - a_4 \sin(\theta_4)\dot{\theta_4}^2$

3 Dynamic Analysis

For Dynamic Analysis, Lagrange method is used [7]. In total there are 4 angles $\theta_1, \theta_2, \theta_3, \theta_4$ and θ_3, θ_4 are the functions of θ_2, θ_5 as shown in Eq.(10).

$$\begin{aligned} \theta_3 &= f(\theta_2, \theta_5) \\ \theta_4 &= g(\theta_2, \theta_3, \theta_5) = g(\theta_2, f(\theta_2, \theta_5), \theta_5) \end{aligned} \tag{10}$$

Position vectors of tip points of each link are given in Eq.(11).

$$\begin{aligned} r_2 &= \begin{bmatrix} \frac{-a_1}{2} + \frac{a_2\cos(\theta_2)}{2} \\ \frac{a_2\sin(\theta_2)}{2} \\ 0 \end{bmatrix} r_3 = \begin{bmatrix} \frac{-a_1}{2} + a_2\cos(\theta_2) + \frac{a_3\sin(\theta_3)}{2} \\ a_2 sin(\theta_2) + \frac{a_3\sin(\theta_3)}{2} \\ 0 \end{bmatrix} \\ r_5 &= \begin{bmatrix} \frac{a_1}{2} + \frac{a_5\cos(\theta_5)}{2} \\ \frac{a_5\sin(\theta_5)}{2} \\ 0 \end{bmatrix} r_4 = \begin{bmatrix} \frac{a_1}{2} + a_5\cos(\theta_5) + \frac{a_4\cos(\theta_4)}{2} \\ a_5\cos(\theta_5) + \frac{a_4\sin(\theta_4)}{2} \\ 0 \end{bmatrix} \end{aligned} \tag{11}$$

We can substitute θ_3 and θ_4 from Eq.(10) as a function of θ_2 and θ_5 for θ_3 and θ_4 in Eq.(11).
Thus, Eq.(12) is obtained as showned below.All position vectors of links depend on the variable θ_2 and θ_5.

$$\begin{aligned} r_2 &= \begin{bmatrix} \frac{-a_1}{2} + \frac{a_2\cos(\theta_2)}{2} \\ \frac{a_2\sin(\theta_2)}{2} \\ 0 \end{bmatrix} r_3 = \begin{bmatrix} \frac{-a_1}{2} + a_2\cos(\theta_2) + \frac{a_3\sin(f(\theta_2,\theta_5))}{2} \\ a_2 sin(\theta_2) + \frac{a_3\sin(f(\theta_2,\theta_5))}{2} \\ 0 \end{bmatrix} \\ r_5 &= \begin{bmatrix} \frac{a_1}{2} + \frac{a_5\cos(\theta_5)}{2} \\ \frac{a_5\sin(\theta_5)}{2} \\ 0 \end{bmatrix} r_4 = \begin{bmatrix} \frac{a_1}{2} + a_5\cos(\theta_5) + \frac{a_4\cos(g(\theta_2,f(\theta_2,\theta_5),\theta_5))}{2} \\ a_5\cos(\theta_5) + \frac{a_4\sin(g(\theta_2,f(\theta_2,\theta_5),\theta_5))}{2} \\ 0 \end{bmatrix} \end{aligned} \tag{12}$$

As a next step, Positional jacobian matrices are calculated.

$$J_{p_{link_i}} = \begin{bmatrix} \frac{\partial r_i}{\partial \theta_2} & \frac{\partial r_i}{\partial \theta_5} \end{bmatrix} \tag{13}$$

$$J_{p_{link_2}} = \begin{bmatrix} \frac{-a_2\sin(\theta_2)}{2} & 0 \\ \frac{a_2\cos(\theta_2)}{2} & 0 \\ 0 & 0 \end{bmatrix} \tag{14}$$

$$J_{p_{link_3}} = \begin{bmatrix} -a_2\sin(\theta_2) - \frac{a_3}{2}\sin(f(\theta_2,\theta_5))\frac{\partial f(\theta_2,\theta_5)}{\partial \theta_2} & \frac{-a_3}{2}\sin(f(\theta_2,\theta_5))\frac{\partial f(\theta_2,\theta_5)}{\partial \theta_5} \\ a_2\cos(\theta_2) + \frac{a_3}{2}\cos(f(\theta_2,\theta_5))\frac{\partial f(\theta_2,\theta_5)}{\partial \theta_2} & \frac{a_3}{2}\cos(f(\theta_2,\theta_5))\frac{\partial f(\theta_2,\theta_5)}{\partial \theta_5} \\ 0 & 0 \end{bmatrix} \tag{15}$$

$$J_{p_{link_4}} = \begin{bmatrix} \frac{-a_4}{2}\sin(\sigma)\frac{\partial \sigma}{\partial \theta_2} & -a_5\sin(\theta_5) + \frac{-a_4}{2}\sin(\sigma))\frac{\partial \sigma}{\partial \theta_5} \\ \frac{a_4}{2}\cos(\sigma)\frac{\partial \sigma}{\partial \theta_2} & a_5\cos(\theta_5) + \frac{a_4}{2}\cos(\sigma)\frac{\partial \sigma}{\partial \theta_5} \\ 0 & 0 \end{bmatrix} \tag{16}$$

where $\sigma = g(\theta_2, f(\theta_2, \theta_5), \theta_5)$

$$J_{p_{link_5}} = \begin{bmatrix} 0 & \frac{-a_5}{2}\sin(\theta_5) \\ 0 & \frac{a_5}{2}\cos(\theta_5) \\ 0 & 0 \end{bmatrix} \tag{17}$$

Calculation of rotation matrices for each link is given in Eq.(19). All links rotate about its own z axis. Eq. (18) demostrates the rotation matrix about z axis by θ.

$$R_z(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{18}$$

$$R_{link_2,z}(\theta) = \begin{bmatrix} \cos(\theta_2) & -\sin(\theta_2) & 0 \\ \sin(\theta_2) & \cos(\theta_2) & 0 \\ 0 & 0 & 1 \end{bmatrix} R_{link_3,z}(\theta) = \begin{bmatrix} \cos(f(\theta_2,\theta_5)) & -\sin(f(\theta_2,\theta_5)) & 0 \\ \sin(f(\theta_2,\theta_5)) & \cos(f(\theta_2,\theta_5)) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$
$$R_{link_4,z}(\theta) = \begin{bmatrix} \cos(\theta_4) & -\sin(\theta_4) & 0 \\ \sin(\theta_4) & \cos(\theta_4) & 0 \\ 0 & 0 & 1 \end{bmatrix} R_{link_5,z}(\theta) = \begin{bmatrix} \cos(\sigma) & -\sin(\sigma) & 0 \\ \sin(\sigma) & \cos(\sigma) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{19}$$

where $\sigma = g(\theta_2, f(\theta_2,\theta_5), \theta_5)$
Orientational jacobian matrix for each link is calculated using Eq.(20).

$$J_{O_{link_i}} = \left[col[\frac{\partial R_{link_i,z}(\theta_i)}{\partial \theta_2} \cdot R^T_{link_i,z}(\theta_i)] \; col[\frac{\partial R_{link_i,z}(\theta_i)}{\partial \theta_5} \cdot R^T_{link_i,z}(\theta_i)] \right] \tag{20}$$

where $col^2 \left[\begin{bmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{bmatrix} \right] = \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix}$

$$J_{O_{link_2}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} \qquad J_{O_{link_3}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \frac{\partial f(\theta_2,\theta_5)}{\partial \theta_2} & \frac{\partial f(\theta_2,\theta_5)}{\partial \theta_5} \end{bmatrix}$$
$$J_{O_{link_4}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \frac{\partial g(\theta_2,f(\theta_2,\theta_5),\theta_5)}{\partial \theta_2} & \frac{\partial g(\theta_2,f(\theta_2,\theta_5),\theta_5)}{\partial \theta_5} \end{bmatrix} J_{O_{link_5}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \tag{21}$$

Dynamic Equation of Motion

$$M(q)\ddot{q} + G(q,\dot{q}) = \tau \tag{22}$$

where Intertia I_{link_i}, Mass $M(q)$,Gyroscope $G(q,\dot{q})$

$$I_{link_i} = \begin{bmatrix} I_{xx_{link_i}} & -I_{xy_{link_i}} & -I_{xz_{link_i}} \\ -I_{xy_{link_i}} & I_{yy_{link_i}} & -I_{yz_{link_i}} \\ -I_{xz_{link_i}} & -I_{yz_{link_i}} & I_{zz_{link_i}} \end{bmatrix} \tag{23}$$

[2] column operator

$$M(q) = \sum_{n=1}^{N} m_{link_i} J_{p_{link_i}}^T J_{p_{link_i}} + J_{O_{link_i}}^T R_{link_i} I_{link_i} R_{link_i}^T J_{O_{link_i}} \quad (24)$$

$$G = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \quad where \quad g_{ij} = \sum_{k=1}^{N} g_{ijk} \dot{q}_k \qquad g_{ijk} = \frac{1}{2}\left(\frac{\partial m_{ij}}{\partial q_k} + \frac{\partial m_{ik}}{\partial q_j} + \frac{\partial m_{jk}}{\partial q_i}\right) \quad (25)$$

4 Dynamic Control

$x = f(q) \quad x \in \mathbb{R}^n$ and $f : \mathbb{R}^n \to \mathbb{R}^n$
Differentiation Forward Kinematic yiels $\dot{x} = J\dot{q}$ where $J(q) \in \mathbb{R}^{nxn}$ is the Jacobian matrix.
$x(t)$, $\dot{x}(t)$, $q(t)$, $\dot{q}(t)$ are available for control design and $M(q)$, $G(q, \dot{q})$, $J(q)$ are known. Let's define $e(t) \in \mathbb{R}^n$ task space tracking error $e = x_d - x$ where $x_d \in \mathbb{R}^n$ is the desired end effector trajectory.
Differentiation velocity kinematic yiels $\ddot{x} = \dot{J}\dot{q} + J\ddot{q}$ so $\ddot{q} = J^{-1}\left(\ddot{x} - \dot{J}\dot{q}\right)$
Design Controller [3]

$$\tau = M(q)J(q)^{-1}\left(\ddot{x}_d + K_v\dot{e} + K_p e - \dot{J}\dot{q}\right) + G(q, \dot{q}) \quad (26)$$

where K_p, $K_n \in \mathbb{R}^{nxn}$ are diagonal,positive definite control gain matrices.

4.1 Speech Command Recognition using Deep Learning

This part shows how to create a simple deep learning model and recognize the speech command.Labelled dataset is downloaded from TensorFlow[3]. The dataset includes recorded wav files which is recorded from various people. Recorded speeches are labelled as ["*zero*","*one*","*two*","*three*",, "*four*","*five*","*six*","*seven*","*eight*","*nine*"]. So the neural network is trained to recognize only these speech commands. Speeches except for specified given as a speech command will be recognized as "unknown" command by the trained network.

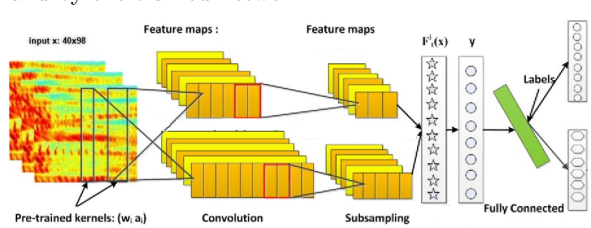

Fig. 3. General network structure for speech recognition [5]

The General structure of the network for speech recognition is shown in the Fig.3. Using **CNN** architecture for this task is demostrated in the Fig.5.In total, 24 layers are used.

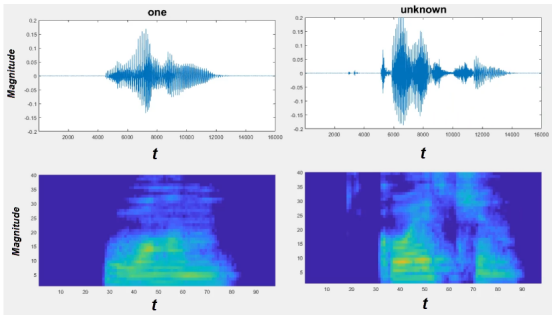

Fig. 4. Waveforms and Spectrograms of my own voices for in recognition process.The word "one" is given as a speech and network recognized successfully.Any word except for trained word is given as a speech. Since the network wasn't trained by this labeled word, It gives a message as "unknown" to indicate that the given speech could not be recognized.

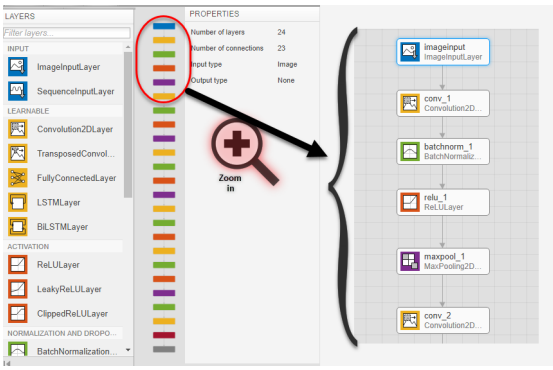

Fig. 5. The used **CNN** architecture

Fully simplified used architrecture are listed below.

[3] $http://download.tensorflow.org/data/speech_commands_v0.01.tar.gz$

Fig. 6. Distribution of the different class labels in the training and validation sets

[$40x98x1$] **INPUT**
[$40x98x12$] CONV1 : 12 3x3 filters at stride 1 ,pad 0
[$40x98x12$] NORM1
[$40x98x12$] RELU1
[$20x49x12$] MAXPOOL1 : 3x3 filters at stride 2 ,pad 0
[$20x49x24$] CONV2 : 24 3x3 filters at stride 1 ,pad 0
[$20x49x24$] NORM2
[$20x49x24$] RELU2
[$10x25x24$] MAXPOOL2 : 12 3x3 filters at stride 2 ,pad 0
[$10x25x48$] CONV3 : 48 3x3 filters at stride 1 ,pad 0
[$10x25x48$] NORM3
[$10x25x48$] RELU3
[$5x13x48$] MAXPOOL3 : 12 3x3 filters at stride 2 ,pad 0
[$5x13x48$] CONV4 : 48 3x3 filters at stride 1 ,pad 0
[$5x13x48$] NORM4
[$5x13x48$] RELU4
[$5x13x48$] CONV5 : 48 3x3 filters at stride 1 ,pad 0
[$5x13x48$] NORM5
[$5x13x48$] RELU5
[$5x1x48$] MAXPOOL4 : 1x13 filters at stride 1 ,pad 0
[$5x1x48$] DROPOUT : 20 %
[$1x1x12$] FC : 12 fully connected layer
[$1x1x12$] SOFTMAX
[$1x12$] OUTPUT

Results of trained network
About 32000 data is used.80% (25000) of it is used for traning , 10% of it is used for validation and the rest is used for test.
Training error: 1.3632% and Validation error: 3.885%
Confusion Matrix is shown in Fig.7, which is also known as error matrix.

5 Simulation

Simulation is run in Matlab/Simulink environment.Controller gain matrices K_p and K_d are selected $\begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$, $\begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$ respectively.
Link lengths are taken from the the paper *(Dede & Kiper)* [2] and are shown in the table 1. For links ,hollow carbon fiber links are taken into account. Determined mass and inertia tensors are illustrated in table 1.

$$I_{link2} = I_{link3} = I_{link4} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} kgm^2, \; I_{link5} = \begin{bmatrix} 1.3 & 0 & 0 \\ 0 & 1.3 & 0 \\ 0 & 0 & 1.3 \end{bmatrix} kgm^2$$

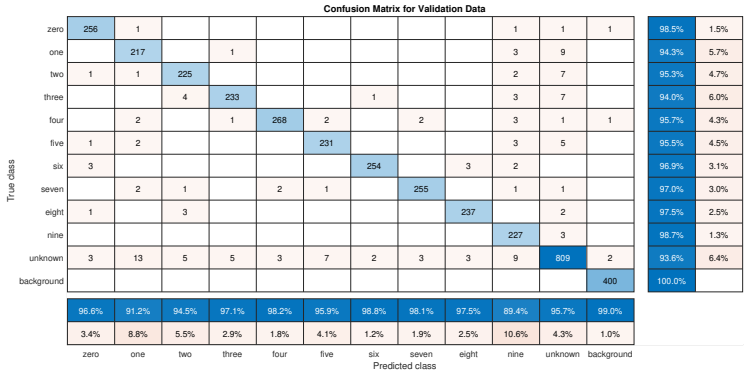

Fig. 7. Confusion Matrix

Table 1. Designed link lengths

a_1	a_2	a_3	a_4	a_5
250mm	250mm	250mm	250mm	200mm
m_1	m_2	m_3	m_4	m_5
0.85kg	0.85kg	0.85kg	0.85kg	0.75kg

The desired trajectory followed by the mechanism is given as input and depends on the recognized speech command. For example, desired input in the case for recognized number 2 is demonstrated in Fig.10.

5.1 Results

After the simulation is executed, the acquired results are given below. Because actuators cannot provide unlimited torque, torque is limited between $+3\ Nm$ and $-3\ Nm$. Torque with saturation block and without saturation block are shown in the Fig.8 and Fig.9. Since results aren't much different between Torque with saturation and without saturation, results shown below are calculated with regard to Torque without saturation.

The curve over number 2 in Fig.11 is due to initial conditions of the mechanism.

It can be seen from Fig.12 that the error value fluctuates between -0.01 and 0.01. In other words, given the desired trajectory in Fig.10 is followed by the end effector of 5R mechanism deviating given trajectory with the amount of error shown in Fig.12.

Fig.13 shows that changing angles of each links while the 5R mechanism follows the given trajectory. Fig.14 illustrates the angular velocity of link 2 and link 5 which are driven by actuators. Since the velocity curve in Fig.14 has no abrupt change, angular acceleration is smooth

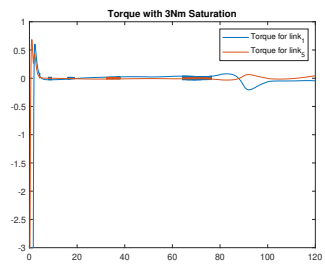

Fig. 8. Torque with saturation $\pm 3Nm$

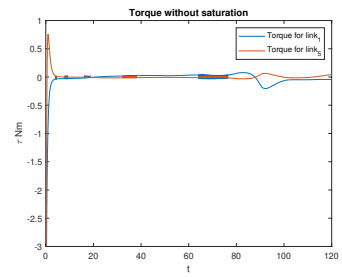

Fig. 9. Torque without saturation

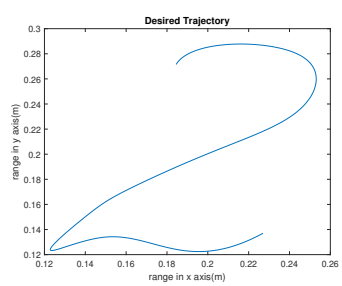

Fig. 10. Desired trajectory for #2

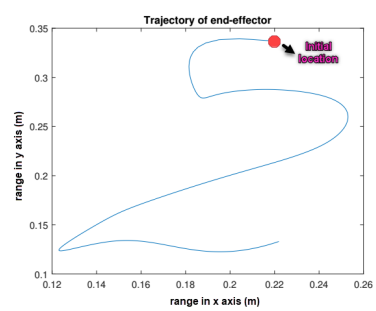

Fig. 11. Trajectory of end-effector

except for initial movement shown in Fig.15. Fig.16 supports that the mechanism does not step in singularity configuration while following the desired trajectory because the determinant of the Jacobian Matrix does not becomes equal to the zero.

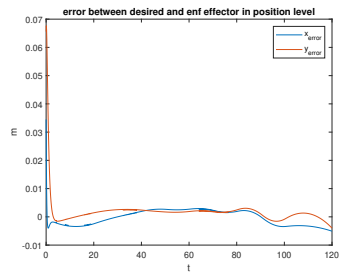

Fig. 12. Task space tracking error $e = x_d - x$.As the mechanism operates in 2D plane, x_{error} and y_{error} mean the error in x axis and y axis.

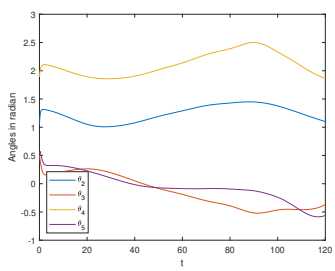

Fig. 13. Changing angles of each links

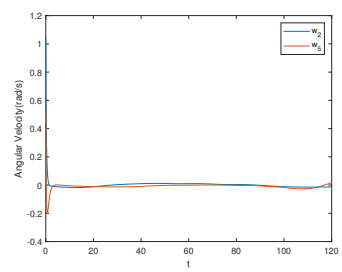

Fig. 14. Changing anglular velocity of link1 and link5

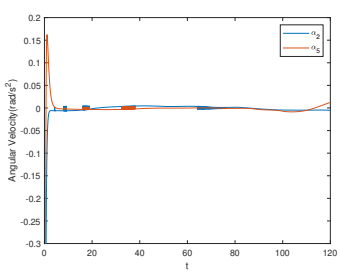

Fig. 15. Changing anglular acceleration of link1 and link5

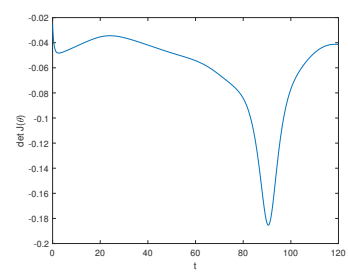

Fig. 16. Determinant of Jacobian Matrix

6 Conclusions

The idea of using handwriting haptic device with deep learning proposed successfully. The proposed model with deep learning has been performed within simulation environment in Matlab/Simulink. Obtained results have been indicated.Based on the results, the idea can be actu-

alized in real life application and it will give great benefit to the people and make their life easier.

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References

1. Coles, T.R., Meglan, D., John, N.W.: The role of haptics in medical training simulators: A survey of the state of the art. IEEE Transactions on Haptics **4**(1), 51–66 (2011). DOI 10.1109/TOH.2010.19
2. Dede, M.İ.C., Kiper, G.: Mechanism design for haptic handwriting assistance device. In: Mechanisms, Transmissions and Applications, pp. 271–278. Springer (2015)
3. Hsu, P., Mauser, J., Sastry, S.: Dynamic control of redundant manipulators. Journal of Robotic Systems **6**(2), 133–148 (1989)
4. Liu, X., Dodds, G., McCartney, J., Hinds, B.: Virtual designworks-designing 3d cad models via haptic interaction. Computer-Aided Design **36**(12), 1129–1140 (2004). DOI 10.1016/j.cad.2003.10.003
5. Mao, Q., Dong, M., Huang, Z., Zhan, Y.: Learning salient features for speech emotion recognition using convolutional neural networks. IEEE Transactions on Multimedia **16**, 2203–2213 (2014)
6. Minogue, J., Jones, M.G.: Haptics in education: Exploring an untapped sensory modality. Review of Educational Research **76**(3), 317–348 (2006). DOI 10.3102/00346543076003317. URL https://doi.org/10.3102/00346543076003317
7. Siciliano, B., Sciavicco, L., Villani, L., Oriolo, G.: Robotics: modelling, planning and control. Springer Science & Business Media (2010)
8. Tarwala, R., Dorr, L.D.: Robotic assisted total hip arthroplasty using the mako platform. Current reviews in musculoskeletal medicine **4**(3), 151 (2011)
9. Yu, W., Ramloll, R., Brewster, S.: Haptic graphs for blind computer users. In: S. Brewster, R. Murray-Smith (eds.) Haptic Human-Computer Interaction, pp. 41–51. Springer Berlin Heidelberg, Berlin, Heidelberg (2001)